CS305 Database Systems
Lecture 18: SQL
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Outline
- Relational model
- Keys
- SQL queries
- Joins
- Normalization
- Transactions

The Relational Model

- A relation is a table of rows and columns.
- Each row is a tuple representing one entity.
- Each column is an attribute with a domain.

Primary Key
- A primary key uniquely identifies each row in a table.
- It cannot contain NULL values.
- Every table should declare one.

Foreign Key
- A foreign key references the primary key of another table.
- It enforces referential integrity between relations.

SELECT Statement
- SELECT retrieves rows from one or more tables.
- The WHERE clause filters rows by a predicate.

Projection
- Projection chooses which columns appear in the result.
- It is written as the column list after SELECT.

Aggregate Functions

- Aggregate functions compute a single value over many rows.
- COUNT returns the number of rows in a group.

GROUP BY
- GROUP BY partitions rows into groups sharing a value.
- Aggregates are then computed per group.

HAVING
- HAVING filters groups after aggregation.
- WHERE filters rows before grouping.

Inner Join
- An inner join returns rows matching in both tables.
- The join condition compares a foreign key to a primary key.

Outer Join
- A left outer join keeps unmatched rows from the left table.
- Missing values are padded with NULL.

Subqueries
- A subquery is a query nested inside another query.
- A correlated subquery references the outer query.

Views
- A view is a named query stored in the schema.
- It presents derived data as if it were a table.

Figure 3: ER diagram

Indexes

- An index is a data structure that speeds up lookups.
- It trades extra write cost for faster reads.

Functional Dependency
- A functional dependency means one attribute determines another.
- It is written X arrow Y.

First Normal Form
- First normal form requires every attribute to be atomic.
- Repeating groups are not allowed.

Second Normal Form
- Second normal form removes partial dependencies on a composite key.
- Every non-key attribute depends on the whole key.

Third Normal Form
- Third normal form removes transitive dependencies.
- Non-key attributes must not determine other non-key attributes.

Denormalization
- Denormalization reintroduces redundancy to improve read speed.
- It is a deliberate trade-off against update anomalies.

Transactions

- A transaction is a unit of work executed atomically.
- It either commits entirely or aborts entirely.

ACID Properties
- Atomicity means all operations succeed or none do.
- Durability means committed changes survive failure.

Isolation Levels
- Isolation determines how concurrent transactions see each other.
- Serializable is the strictest level.

Deadlock
- A deadlock occurs when transactions wait on each other forever.
- The system resolves it by aborting one transaction.

Stored Procedures
- A stored procedure is program logic stored in the database.
- It reduces round trips between client and server.

Triggers

- A trigger executes automatically in response to a table event.
- It can enforce rules that constraints cannot express.

Query Optimization
- The optimizer chooses a low cost execution plan.
- Cost is estimated from table statistics.

Figure 7: query plan tree

Summary
- Keys, joins, normalization, transactions

References
Silberschatz, Database System Concepts

Appendix
- Additional reading